SYNOPSIS

1.1 Project Title and Group

Field	Example
Project Title	Smart Expense & Budget Management Application
Group Number	G-15
Project ID	15
Customer / Stakeholder	Students/Working professionals.

1.2 Team Profile and Product Ownership

Member	SRN	Owned functionality	Secondary responsibilities
Member 1 Rhea Sushanth	PES1UG24CS371	Budget Setting, Alerts & Reporting/Analytics	CI/CD Pipeline
Member 2 Sarthki Vachhani	PES1UG24CS422	User Auth & Profile Management	UI/UX, Testing
Member 3 Sai Manogna BV	PES1UG24CS399	Expense/Income Transaction Management	API Design Review

Testing can be distributed(among 3), each member tests their own owned feature

1.3 Problem Statement

Most individuals track personal expenses inconsistently through scattered notes, memory, or generic spreadsheets leading to poor visibility into spending patterns and little to no proactive budget control. Existing solutions are often either too complex (full accounting suites) or too simplistic (plain expense loggers with no budgeting or analytics). The Smart Expense & Budget Management Application aims to give users a lightweight, structured way to log expenses, set category-wise budgets,

and get real-time visibility into their financial habits.

1.4 Intended Users and Scope

Intended Users

- Students managing limited monthly allowances/pocket money who need quick expense logging and budget alerts.
- Working professionals tracking discretionary spending, recurring bills, and monthly savings targets.
- Administrators (project-level) managing expense categories and basic user oversight.

Scope (In)

- Single-user personal finance tracking (not multi-user shared households/accounts).
- Manual and CSV-based transaction entry; sandbox/dummy bank-linking as a stretch feature only.
- Category-wise budgeting, threshold alerts, and analytics/reporting dashboards.

Scope (Out)

- Real bank account linking or live payment processing.
- Multi-currency conversion or investment/portfolio tracking.
- Multi-user shared budgets (e.g., family accounts) noted as a possible future enhancement

1.5 Objectives

- Provide a simple, fast way to record and categorize daily expenses.
- Allow users to set budgets per category/time period and track adherence.
- Give users visual insight into spending trends through reports and dashboards.
- Alert users when spending approaches or exceeds a set budget.
- Provide administrators with basic category/user management capability.

- Develop the product incrementally using Agile practices.

1.6 Functional Features

1. User registration and login.
2. Expense entry and categorization.
3. Budget creation and tracking (per category/period).
4. Transaction import (manual/CSV, with sandbox bank-linking as a stretch feature).
5. Spending reports and analytics dashboard.
6. Budget threshold alerts/notifications.
7. Recurring transaction handling.
8. Admin/category management.

1.7 Short-Term Plan

Iteration	Planned Output	Acceptance Evidence
Sprint 1	Project setup, requirements baseline, login	User stories + Git repo + working login
Sprint 2	Expense entry, categorization	Demo + unit tests
Sprint 3	Budget creation & tracking logic	Integration tests + PRs
Sprint 4	Reports/analytics dashboard	System demo + test evidence
Sprint 5	Notifications, admin panel, security hardening	Validation + coverage report
Sprint 6	System validation and release	Final report + deployment/demo

1.8 Proposed Technology / Tool Stack

Backend: Python (Django)

Database: PostgreSQL / SQLite (dev)

Frontend: Django templates + Chart.js (for reports/analytics) + HTMX

Version Control: Git + GitHub

Testing: Pytest / Django test framework

CI: GitHub Actions

1. 9 Agile Justification

Agile (Scrum-style sprints) fits this project because requirements around budgeting/reporting features are likely to evolve as user feedback comes in during demos. Short, fixed-length sprints (as laid out above) let the team deliver working increments early (starting with login in Sprint 1), gather feedback, and adjust scope e.g., deciding how deep to go on bank-linking without blocking the whole project on upfront design of every feature.